Compound Angle Formulae

Exam code: 9MA0

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Written by: Roger B
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Compound angle formulae

What are the compound angle formulae?

- There are six **compound angle formulae** (also known as **addition formulae**), two each for **sin**, **cos** and **tan**:
- For **sin** the +/- sign on the left-hand side **matches** the one on the right-hand side

$$\sin(A+B) \equiv \sin A \cos B + \cos A \sin B$$
$$\sin(A-B) \equiv \sin A \cos B - \cos A \sin B$$

- For **cos** the +/- sign on the left-hand side is **opposite to** the one on the right-hand side

$$\cos(A+B) \equiv \cos A \cos B - \sin A \sin B$$
$$\cos(A-B) \equiv \cos A \cos B + \sin A \sin B$$

- For **tan** the +/- sign on the left-hand side **matches** the one in the **numerator** on the right-hand side, and is **opposite to** the one in the **denominator**

$$\tan(A+B) \equiv \frac{\tan A + \tan B}{1 - \tan A \tan B}$$
$$\tan(A-B) \equiv \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

- You can **derive** the tan identity by:

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$\cos A \cos B$

Examiner Tips and Tricks

- All these formulae are in the formulae booklet – you don't have to memorise them.

Worked Example

a) Express $\tan(225° - 30°)$ in terms of $\tan 225°$ and $\tan 30°$.

b) Hence show that $\tan 195° = 2 - \sqrt{3}$.

a) $\tan(A-B) \equiv \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$ ← FORMULA 6 ABOVE

$\tan(225° - 30°) \equiv \dfrac{\tan 225° - \tan 30°}{1 + \tan 225° \tan 30°}$

b) $\tan 225° = 1$ $\quad \tan 30° = \dfrac{\sqrt{3}}{3}$ ← YOU SHOULD KNOW THESE!

$\tan(195°) = \tan(225° - 30°)$

$= \dfrac{1 - \frac{\sqrt{3}}{3}}{1 + (1)(\frac{\sqrt{3}}{3})} = \dfrac{1 - \frac{\sqrt{3}}{3}}{1 + \frac{\sqrt{3}}{3}}$

$\left[\times \frac{3}{3}\right] = \dfrac{3 - \sqrt{3}}{3 + \sqrt{3}}$

$= \dfrac{3 - \sqrt{3}}{3 + \sqrt{3}} \times \dfrac{3 - \sqrt{3}}{3 - \sqrt{3}}$ ← RATIONALISE THE DENOMINATOR

$= \dfrac{12 - 6\sqrt{3}}{6}$

$= 2 - \sqrt{3}$

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